

End-to-End HuggingFace Model Training & Docker Deployment for Genre Classification

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Model: https://huggingface.co/spidey1807/mlops_ass3

Github: <https://github.com/ShubhamHaraniya/MLOps-Shubham-M25CSA013/tree/Assignment-3>

Abstract: I fine-tune distilbert-base-uncased (66M parameters) on the UCSD Goodreads dataset for 8-genre book review classification. The model achieves **61.88% accuracy** (F1: 0.6191), a $\sim 5\times$ improvement over the random baseline. The pipeline includes training, evaluation with visualizations, HuggingFace Hub deployment, and GPU-enabled Docker containerization. Local and Hub evaluations produce identical results, validating deployment integrity.

1. Introduction

In this project, I implement a complete MLOps pipeline: (1) stream book reviews from UCSD Goodreads (8 genres, 16K samples), (2) fine-tune a DistilBERT model, (3) evaluate with metrics and visualizations, (4) push to HuggingFace Hub, (5) re-evaluate from Hub to verify integrity, and (6) containerize with GPU-enabled Docker. Target genres: *Children, Comics & Graphic, Fantasy & Paranormal, History & Biography, Mystery/Thriller/Crime, Poetry, Romance, Young Adult*.

2. Model Selection

I chose distilbert-base-uncased for the following reasons:

- **Knowledge Distillation:** DistilBERT retains 97% of BERT's capabilities while being 60% faster and 40% smaller (66M vs 110M parameters) [1].
- **Uncased Variant:** Genre classification depends on semantics, not capitalization.
- **Compatibility:** Natively supports AutoModelForSequenceClassification for our 8-class task.
- **Efficiency:** Fine-tunes on a single GPU in ~ 20 minutes.

3. Training Summary

Dataset: UCSD Goodreads Book Graph [2] — 10K reviews streamed per genre, 2K sampled, split 800/200 per genre = **6,400 train + 1,600 test**.

Table 1: Training Configuration & Results

Parameter	Value	Result	Value
Epochs	5	Final Loss	0.7048
Batch Size	8 / 16	Runtime	1236.3s (~ 20 min)
Learning Rate	5×10^{-5}	Throughput	25.88 samples/s
Warmup Steps	100	Total Steps	4,000
Weight Decay	0.01	Total FLOPs	2.12×10^{15}
Max Length	256	Optimizer	AdamW

4. Evaluation Results

4.1 Overall Metrics — Local vs. Hub

Table 2: Local vs. Hub Model Evaluation (zero difference confirms correct deployment)

Metric	Local	Hub	Diff
Accuracy	0.6188	0.6188	0.0000
Precision	0.6202	0.6202	0.0000
Recall	0.6188	0.6188	0.0000
F1-Score	0.6191	0.6191	0.0000
Loss	1.6575	1.6575	0.0000

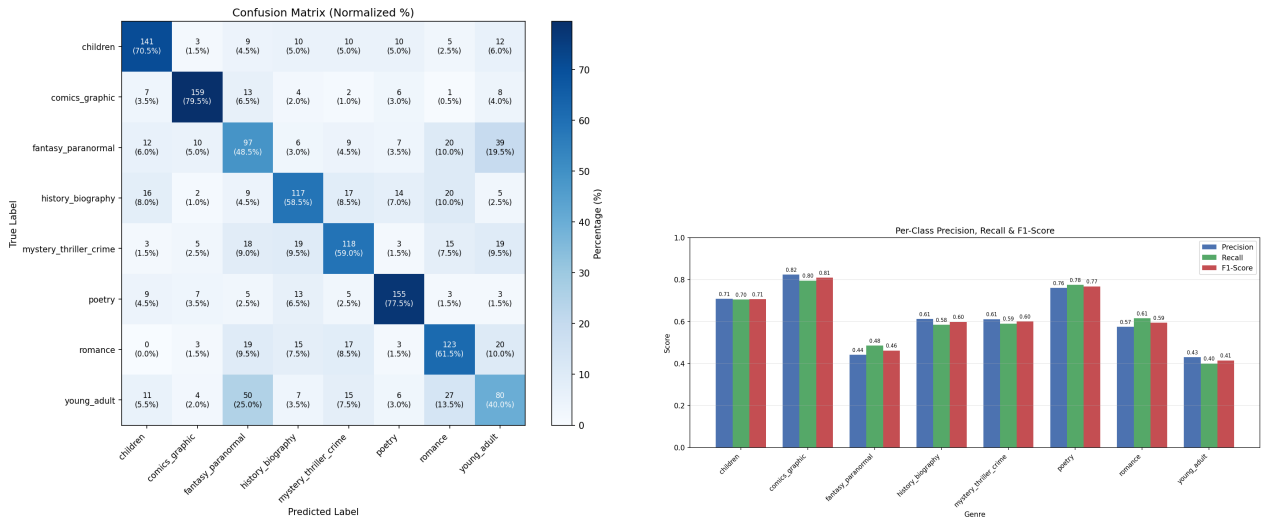
4.2 Per-Class Performance

Table 3: Per-Class Classification Report

Genre	Precision	Recall	F1	N
Children	0.71	0.70	0.71	200
Comics & Graphic	0.82	0.80	0.81	200
Fantasy & Paranormal	0.44	0.48	0.46	200
History & Biography	0.61	0.58	0.60	200
Mystery/Thriller/Crime	0.61	0.59	0.60	200
Poetry	0.76	0.78	0.77	200
Romance	0.57	0.61	0.59	200
Young Adult	0.43	0.40	0.41	200
Macro Avg	0.62	0.62	0.62	1600

Comics & Graphic (F1: 0.81) and Poetry (F1: 0.77) perform best due to distinctive vocabulary. Young Adult (0.41) and Fantasy (0.46) are the weakest due to vocabulary overlap with other fiction genres.

4.3 Visualizations



(a) Confusion Matrix (Normalized %)

(b) Per-Class Precision, Recall, F1

Figure 1: Classification analysis visualizations.

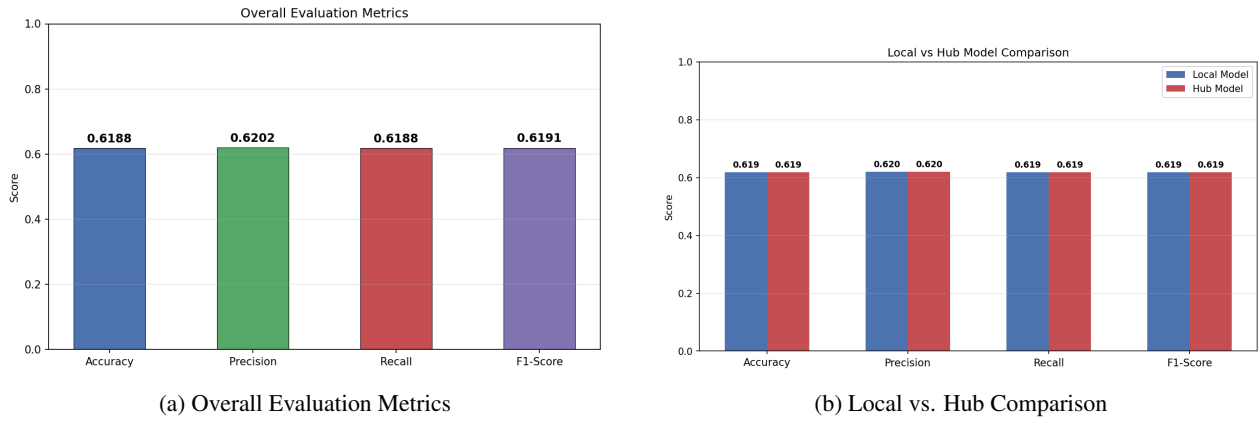


Figure 2: Overall metrics and deployment verification.

5. Docker Deployment

Development Image (Dockerfile): Based on `pytorch/pytorch:2.5.1-cuda12.4-cudnn9-runtime`. Supports the full pipeline with GPU via `-gpus all`.

Production Image (Dockerfile.eval): GPU-enabled, downloads model from HuggingFace Hub on startup. Configured via `HF_TOKEN` and `HF_REPO` environment variables.

```
# Full pipeline          | # Production eval
docker run --gpus all    | docker run --gpus all \
  mlops-ass3 python      |   -e HF_TOKEN=<token> \
  main.py --all          |   -e HF_REPO=spidey1807/mlops_ass3 \
  --repo_name ...        |   mlops-ass3-eval
```

6. Challenges

- Dependency Conflicts:** NumPy 2.x incompatible with PyTorch/Transformers — resolved by pinning `numpy<2.0.0`.
- GPU in Docker:** Required switching to official PyTorch CUDA base image and `-gpus all`.
- Data Latency:** Streaming 80K reviews caused network delays — used efficient HTTP streaming.
- Genre Overlap:** Young Adult and Fantasy overlap with fiction genres, limiting performance.
- Reproducibility:** Random sampling without fixed seed causes slight run-to-run variation.

7. Conclusion

I successfully implemented an end-to-end MLOps pipeline achieving **61.88% accuracy** on 8-genre classification ($\sim 5\times$ over random baseline). Identical local/hub results validate deployment integrity. The project is containerized with GPU-enabled Docker and publicly available on HuggingFace Hub.

References

- [1] V. Sanh, L. Debut, J. Chaumond, T. Wolf, “DistilBERT, a distilled version of BERT: smaller, faster, cheaper and lighter,” *arXiv:1910.01108*, 2019.
- [2] M. Wan, J. McAuley, “Item recommendation on monotonic behavior chains,” *Proc. ACM RecSys*, 2018.